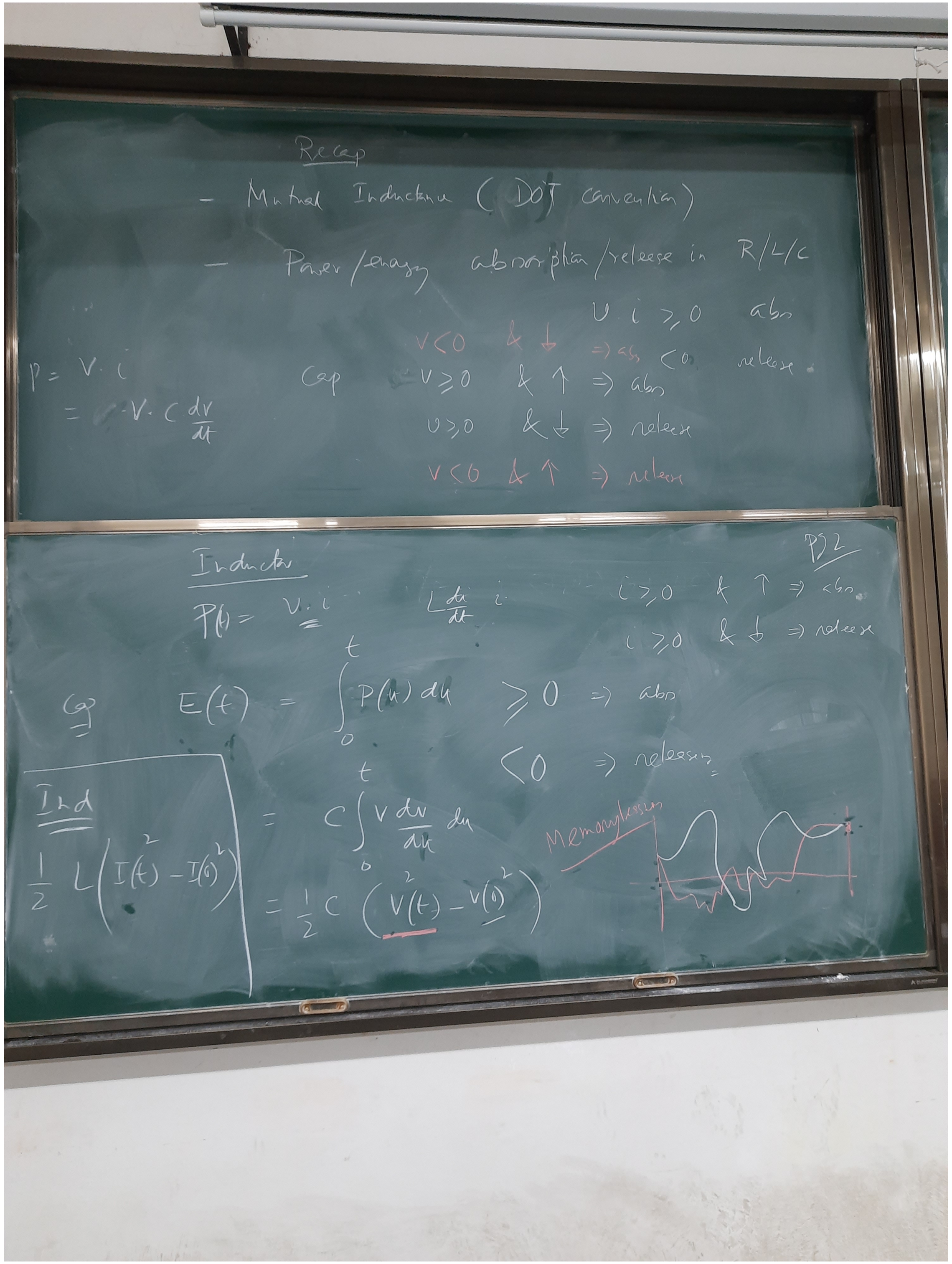


Recap

- Mutual Inductance (DOT convention)
- Power/energy absorption/release in R/L/C

$v \cdot i \geq 0$ abs

$v < 0$ & ↓ ⇒ abs < 0 release

$p = v \cdot i$

$= v \cdot C \frac{dv}{dt}$

Cap $v \geq 0$ & ↑ ⇒ abs

$v \geq 0$ & ↓ ⇒ release

$v < 0$ & ↑ ⇒ release

PS2

Inductor

$P(t) = v \cdot i$ $L\frac{di}{dt} i$

$i \geq 0$ & ↑ ⇒ abs

$i \geq 0$ & ↓ ⇒ release

Cap

$$E(t) = \int_0^t P(u)\,du$$

≥ 0 ⇒ abs

< 0 ⇒ releasing

$$= C\int_0^t v\frac{dv}{du}\,du$$

$$= \frac{1}{2}C\left(V(t)^2 - V(0)^2\right)$$

Ind

$$\frac{1}{2}L\left(I(t)^2 - I(0)^2\right)$$

Memoryless

Inductor

PS2

$P(t) = v \cdot i \qquad L\frac{di}{dt}\, i$

$i \geq 0$ & ↑ ⇒ abs

$i \geq 0$ & ↓ ⇒ release

Cap

$$E(t) = \int_0^t P(u)\,du$$

≥ 0 ⇒ abs

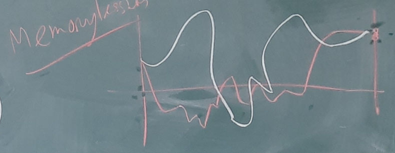
< 0 ⇒ releasing

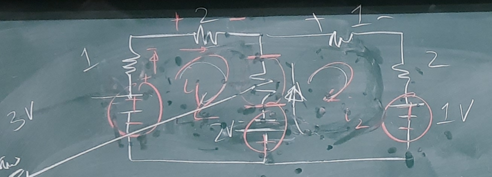
$$= C\int_0^t v\frac{dv}{du}\,du$$

$$= \frac{1}{2}C\left(V(t)^2 - V(0)^2\right)$$

Ind

$$\frac{1}{2}L\left(I(t)^2 - I(0)^2\right)$$





Ohm's law w/ eff conv

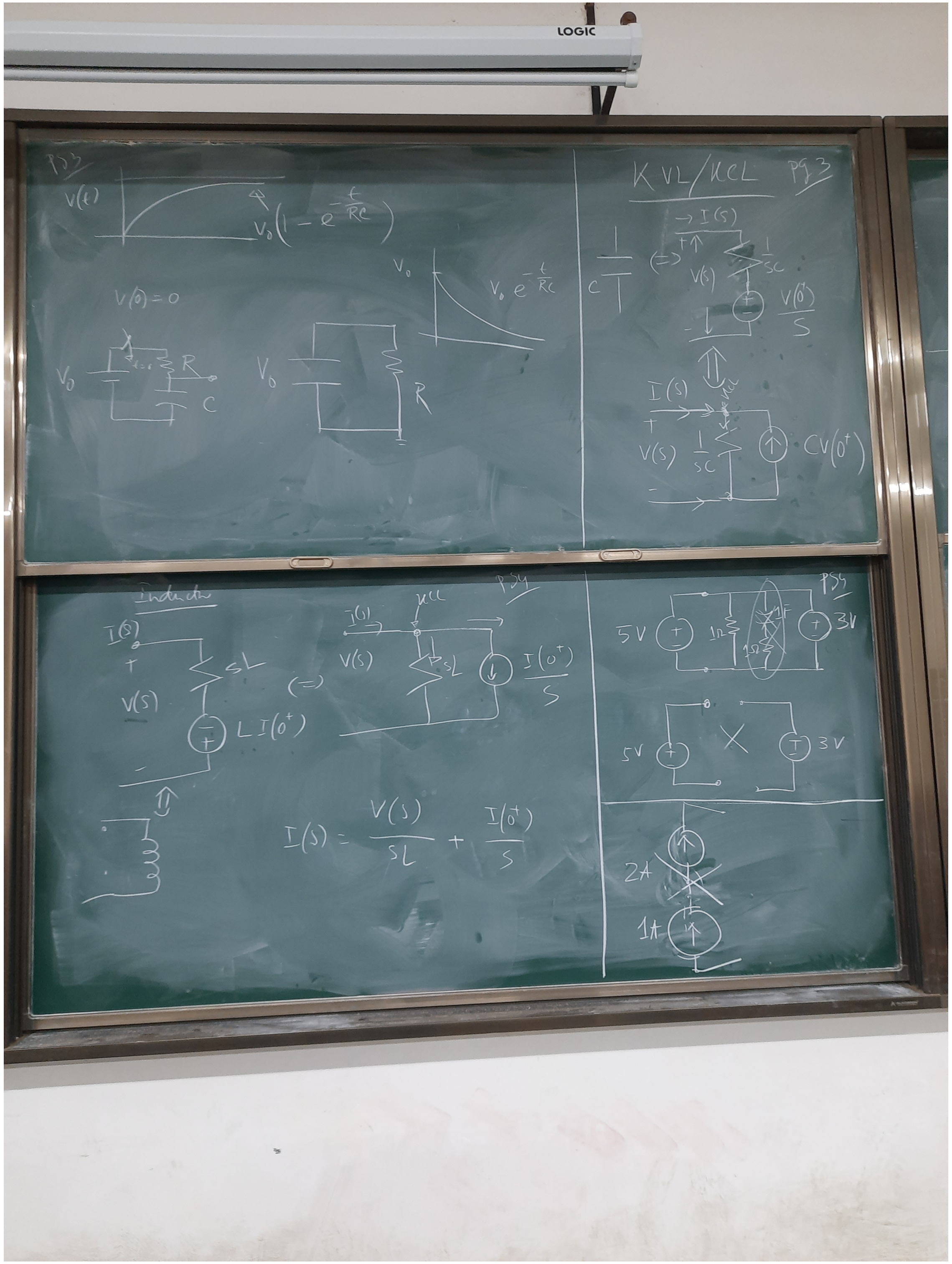
KVL ⟺ Mesh analysis + "mesh-current"

Go for smallest loops

Mesh → that do NOT contain any other loops!

$$-3 + i_1 1 + i_1 2 + (i_1 - i_2)\cdot 1 + -2V = 0$$

$$+2 + (i_2 - i_1)\cdot 1 + i_2 \cdot 1 + i_2 \cdot 2 + 1 = 0$$



$$I(s) = \frac{V(s)}{sL} + \frac{I(0^+)}{s}$$

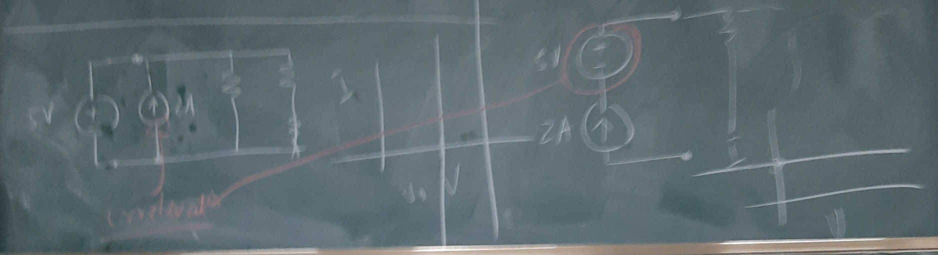
PSS

Voltage source par ✗

series ✓

Curr source par ✓

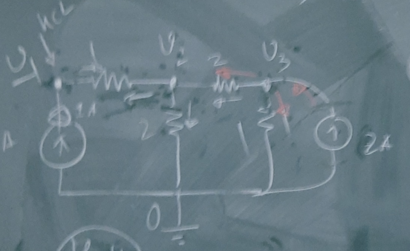
series ✗



Nodal Analysis

↳ use KCL w/ "node-voltage"

(hidden: Ohm's law ⇒ V/I linear relationship)



{ Take current direction & be consistent w/ pot drop

KCL

node 1: $1A = \boxed{\frac{v_1 - v_2}{1}}$

$$\frac{v_2 - v_1}{1} + \frac{v_2 - 0}{2} = \frac{v_3 - v_2}{2}$$

$$\frac{v_3 - v_2}{2} + \frac{v_3 - 0}{1} = 2$$